

# PCA - Principal Components Analysis

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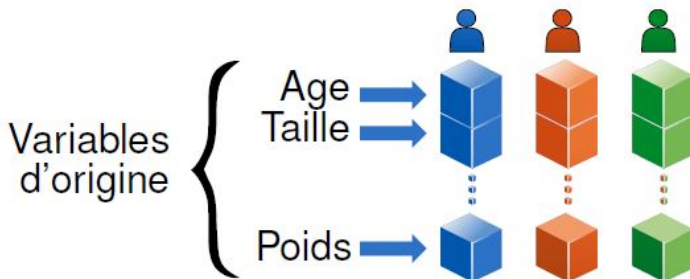


- 1 What is PCA?
- 2 Means, Covariances and Co
- 3 Eigenvectors, Eigenvalues and Matrix Diagonalization
- 4 PCA in Action
- 5 Illustration

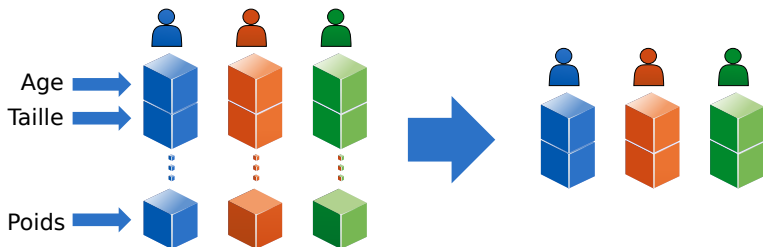


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How to represent data in a space with more than 3 dimensions?



## Idea

Representing data in 2D or 3D with the least possible loss of information as possible.

## principles

- Reduce the number of variables considered
- Minimising loss of information

## Interest

Simplify data to improve :

- their visualization
- their understanding
- their generalization (the more variables  $\rightarrow$  the more risk of overlearning)



## Type 1 - Selection of features

The main variables are part of the original variables

- Advantage: easier understanding of the result
- Default: more limited choices

## Type 2 - Feature Projection

The main variables are not original variables

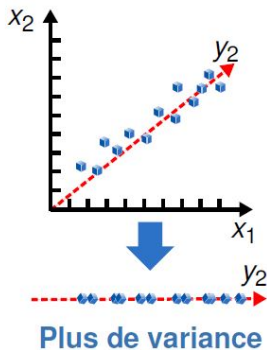
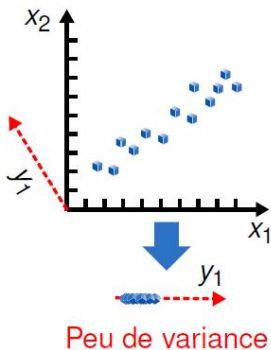
Best-known example: Principal components analysis (PCA)



Size reduction  $\Rightarrow$  loss of information / variance

## Objective

Maximising the variance of the data.



The Principal Component Analysis (PCA) is a method of data analysis which :

- transforms a set of correlated variables into new uncorrelated variables called principal components,
- reduces the significant information using only a smaller number of components (principal) than the original number of variables,
- is an approach that is both
  - geometrical: it is the representation of the variables in a new geometric space in orthogonal directions,
  - statistical: it is looking for best independent new axes which explain the variability - the variance- of the data;
- is used when you want to reduce the number of variables, considering only the first k axes of the PCA in order to "explain" at best the variances of the points considered.



Principal Component Analysis is based on the joint use of two basic tools from statistics and algebra:

- the matrix of covariance (or correlation) of variables expressing their linear relationship,
- the orthogonalization of matrices by spectral decomposition, i.e. using the eigenvectors and eigenvalues.

Reminders...



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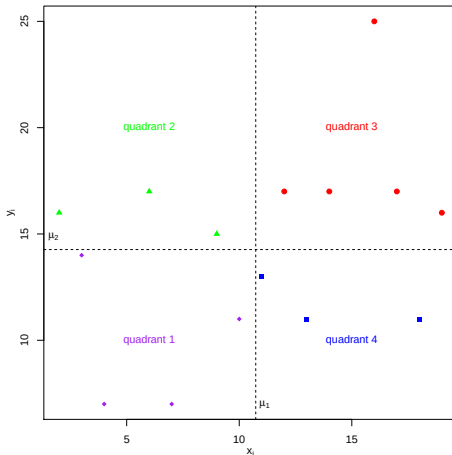


Let us recall:

- The mean :  $\bar{m} = \frac{1}{n} \sum_{i=1}^n x_i$ ,
- The mean deviation :  $s_e(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{m}) = 0 = \text{useless}$ ,
- The real mean deviation :  $s_e(x) = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{m}|$  but it is sh\*t,
- The variance :  $\sigma^2(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{m})^2$  but result in *unit*<sup>2</sup>!
- The standard deviation :  $\sigma(x) = \sqrt{\sigma^2(x)}$ ,
- The covariance  $cov(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{m}_x)(y_i - \bar{m}_y)$

| $x_i$             | $y_i$             | $(x_i - \bar{x})$ | $(y_i - \bar{y})$ | $\times$ of signs | sign of term |
|-------------------|-------------------|-------------------|-------------------|-------------------|--------------|
| 2                 | 16                | $(2 - 10,73)$     | $(16 - 14,26)$    | $- \cdot +$       | $-$          |
| 7                 | 7                 | $(7 - 10,73)$     | $(7 - 14,26)$     | $- \cdot -$       | $+$          |
| 14                | 17                | $(14 - 10,73)$    | $(17 - 14,26)$    | $+ \cdot +$       | $+$          |
| 17                | 17                | $(17 - 10,73)$    | $(17 - 14,26)$    | $+ \cdot +$       | $+$          |
| 3                 | 14                | $(3 - 10,73)$     | $(14 - 14,26)$    | $- \cdot -$       | $+$          |
| 6                 | 17                | $(6 - 10,73)$     | $(17 - 14,26)$    | $- \cdot +$       | $-$          |
| 16                | 25                | $(16 - 10,73)$    | $(25 - 14,26)$    | $+ \cdot +$       | $+$          |
| 18                | 11                | $(18 - 10,73)$    | $(11 - 14,26)$    | $+ \cdot -$       | $-$          |
| 11                | 13                | $(11 - 10,73)$    | $(13 - 14,26)$    | $+ \cdot -$       | $-$          |
| 13                | 11                | $(13 - 10,73)$    | $(11 - 14,26)$    | $+ \cdot -$       | $-$          |
| 10                | 11                | $(10 - 10,73)$    | $(11 - 14,26)$    | $- \cdot -$       | $+$          |
| 12                | 17                | $(12 - 10,73)$    | $(17 - 14,26)$    | $+ \cdot +$       | $+$          |
| 9                 | 15                | $(9 - 10,73)$     | $(15 - 14,26)$    | $- \cdot +$       | $-$          |
| 4                 | 7                 | $(4 - 10,73)$     | $(7 - 14,26)$     | $- \cdot -$       | $+$          |
| 19                | 16                | $(19 - 10,73)$    | $(16 - 14,26)$    | $+ \cdot +$       | $+$          |
| $\bar{x} = 10,73$ | $\bar{y} = 14,26$ |                   |                   |                   |              |

| $x_i$    | 2  | 7 | 14 | 17 | 3  | 6  | 16 | 18 | 11 | 13 | 10 | 12 | 9  | 4 | 19 |
|----------|----|---|----|----|----|----|----|----|----|----|----|----|----|---|----|
| $y_i$    | 16 | 7 | 17 | 17 | 14 | 17 | 25 | 11 | 13 | 11 | 11 | 17 | 15 | 7 | 16 |
| quadrant | 2  | 1 | 3  | 3  | 1  | 2  | 3  | 4  | 4  | 4  | 1  | 3  | 2  | 1 | 3  |



Let  $X$  an array/matrix of real numbers,  $X \in \mathbb{R}^{n \times p}$

$$X = \begin{pmatrix} x_1^1 & \cdots & x_1^j & \cdots & x_1^p \\ \vdots & & \vdots & & \vdots \\ x_i^1 & \cdots & x_i^j & \cdots & x_i^p \\ \vdots & & \vdots & & \vdots \\ x_n^1 & \cdots & x_n^j & \cdots & x_n^p \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_i \\ \vdots \\ \mathbf{x}_n \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{x}^1 & \cdots & \mathbf{x}^j & \cdots & \mathbf{x}^p \end{pmatrix}$$

- $x_i^j$  is the value of the  $i^{th}$  line and of the  $j^{th}$  column,
- $\mathbf{x}^j$  is the  $j^{th}$  column,
- $\mathbf{x}_i$  is the  $i^{th}$  line/vector.

Suppose you want to calculate the averages of the variables:

$$\forall j, \overline{m^j} = \frac{1}{n} \sum_{i=1}^n x_i^j = \sum_{i=1}^n \frac{1}{n} \cdot x_i^j$$

A weighted average of the variables can be determined by giving a weight  $w_1, w_2, \dots$  to the variables (with  $\sum_{i=1}^n w_i^j = 1$ ):

$$\forall j, \overline{m^j} = \sum_{i=1}^n w_i x_i^j$$



Let's write this in matrix form :

$$\begin{pmatrix} \overline{m^1} \\ \vdots \\ \overline{m^p} \end{pmatrix} = X^t W \mathbf{1} = \begin{pmatrix} x_1^1 & \cdots & x_n^1 \\ \vdots & & \vdots \\ x_1^p & \cdots & x_n^p \end{pmatrix} \begin{pmatrix} w_1 & & \\ & \ddots & \\ & & w_n \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$



We can calculate the centered values/variables as follows

$$\begin{aligned}
 Z &= X - \mathbf{1}_n \cdot \overline{m}^t \\
 &= \begin{pmatrix} x_1^1 & \cdots & x_1^p \\ \vdots & & \vdots \\ x_n^1 & \cdots & x_n^p \end{pmatrix} - \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} (\overline{m}^1, \dots, \overline{m}^p) \\
 &= \begin{pmatrix} z_1^1 & \cdots & z_1^p \\ \vdots & & \vdots \\ z_n^1 & \cdots & z_n^p \end{pmatrix}
 \end{aligned}$$

The weighted variance :  $\sigma^2(x) = \sum_{i=1}^n w_i (x_i - \bar{m})^2$

The weighted variance of the  $j$ th variable:

$$\begin{aligned}\sigma^2(\mathbf{x}^j) &= \sum_{i=1}^n w_i \underbrace{(x_i^j - \bar{m}^j)^2}_{z_i^j} \\ &= \sum_{i=1}^n w_i (z_i^j)^2 \\ &= \sum_{i=1}^n z_i^j w_i z_i^j \\ &= (\mathbf{z}^j)^t \mathbf{W} \mathbf{z}^j\end{aligned}$$

The weighted covariance of  $j$ th et  $k$ th variables:

$$\begin{aligned} cov(\mathbf{x}^j, \mathbf{x}^k) &= \sum_{i=1}^n w_i (x_i^j - \overline{m^j})(x_i^k - \overline{m^k}) \\ &= \sum_{i=1}^n w_i z_i^j z_i^k \\ &= \sum_{i=1}^n z_i^j w_i z_i^k \\ &= (\mathbf{z}^j)^t \mathbf{W} \mathbf{z}^k \end{aligned}$$

The variances-covariances matrix  $S$ :

$$S = Z^t W Z$$

The elements on the diagonal of this matrix give the variance for each variable. Hence, the trace of this matrix gives us the total variance of the variables.

$$Tr(S) = \sum_{j=1}^p \sigma^2(\mathbf{x}^j)$$



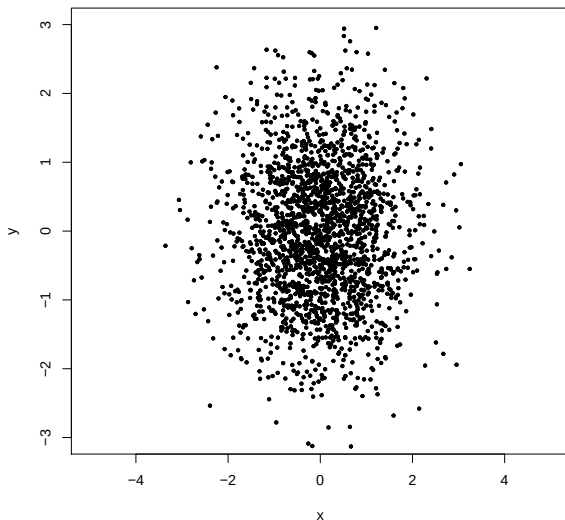
Let  $x_i \approx \mathcal{N}(0, 1)$ ,  $p$  uncorrelated variables and  $X$  coordinates of an individual:

$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_p \end{pmatrix} = S \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix}$$

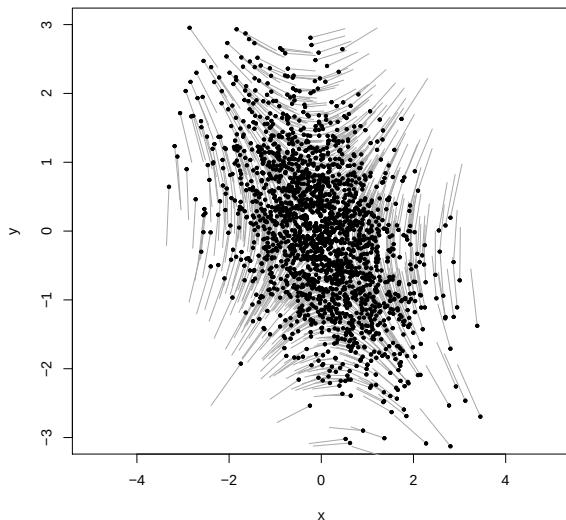
Then  $Y$  is the coordinates of an individual where  $y_i \approx \mathcal{N}(0, 1)$  are  $p$  correlated variables with variance-covariance matrix  $S$ .



## Uncorrelated variables



## Correlated variables



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Let  $V = \mathbb{R}^n$  the real vector space of dimension  $n$ .

If  $\{\vec{b}_1, \dots, \vec{b}_n\}$  is a basis of  $V$ , then each vector  $\vec{v}$  can be written as a linear combination of the  $\vec{b}_i$  vectors:

$$\vec{v} = (x_1, \dots, x_n) = \sum_{i=1}^n a_i \cdot \vec{b}_i$$

Then the coordinates of  $\vec{v} = (x_1, \dots, x_n)$  in the basis  $\{\vec{b}_1, \dots, \vec{b}_n\}$  are simply given by

$$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$$

If  $\{\vec{b}_1, \dots, \vec{b}_n\}$  is the canonical basis, then it is very simple :  $a_i = x_i$ ,  $\forall i \in \{1, \dots, n\}$ .



Let  $T : \mathbb{R}^2 \mapsto \mathbb{R}^2$  the following linear transformation of the  $\mathbb{R}^2$  space :

$$T(x, y) = (-2y; y - x).$$

Using the linear property of  $T$ , the image of a vector  $(x, y) \in \mathbb{R}^2$  by  $T$  expressed in the canonical basis, is given by

$$T(x, y) = T(x \cdot \vec{b}_1 + y \cdot \vec{b}_2) = x \cdot T(\vec{b}_1) + y \cdot T(\vec{b}_2).$$

Then the values of  $T(\vec{b}_1)$  and  $T(\vec{b}_2)$  determine completely  $T$ :

$$T(\vec{b}_1) = T(1; 0) = (0; -1),$$

$$T(\vec{b}_2) = T(0; 1) = (-2; 1).$$



Then using the coordinates of  $T(\vec{b}_1)$  and  $T(\vec{b}_2)$  as columns vectors, we can form a matrix  $A$

$$A = \begin{pmatrix} 0 & -2 \\ -1 & 1 \end{pmatrix}.$$

Then the coordinates of  $T(x, y)$  in the canonical basis are given by

$$A.X = \begin{pmatrix} 0 & -2 \\ -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2y \\ y - x \end{pmatrix}$$

Be aware: for another basis  $\{\vec{u}_1, \vec{u}_2\}$  the coordinates of  $(x; y)$  will be different and the matrix representation of  $T$  will give another matrix  $B$ :  
If  $X$  are the coordinates of  $(x; y)$  in the canonical basis and  $Y$  the coordinates of the same point in the former basis, then there exists a change-of-basis matrix  $Q$  such  $X = Q.Y$ , then

$$A.X = A.Q.Y = B.Y \Rightarrow B = A.Q.$$



Note that a given matrix can then be seen as the matrix representation of a linear transformation for the coordinates of vectors in the canonical basis:

$$A = \begin{pmatrix} 0 & -2 \\ -1 & 1 \end{pmatrix}.$$

and then the image of any point  $(x, y)$  can be computed simply by

$$A.X = \begin{pmatrix} 0 & -2 \\ -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}$$



Let the following vectors in  $\mathbb{R}^2$

$$v_1 = \left( \frac{1}{2}, \frac{3}{2} \right), \quad v_2 = \left( -\frac{3}{2}, \frac{1}{2} \right), \quad v_3 = \left( -1, -\frac{3}{2} \right),$$

$$v_4 = (2, -1), \quad v_5 = (-2, -1), \quad v_6 = (-1, 1).$$

these vectors have the following coordinates in the canonical basis :

$$V_1 = \begin{pmatrix} \frac{1}{2} \\ \frac{3}{2} \end{pmatrix} \quad V_2 = \begin{pmatrix} -\frac{3}{2} \\ \frac{1}{2} \end{pmatrix} \quad V_3 = \begin{pmatrix} -1 \\ -\frac{3}{2} \end{pmatrix}$$

$$V_4 = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad V_5 = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \quad V_6 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$



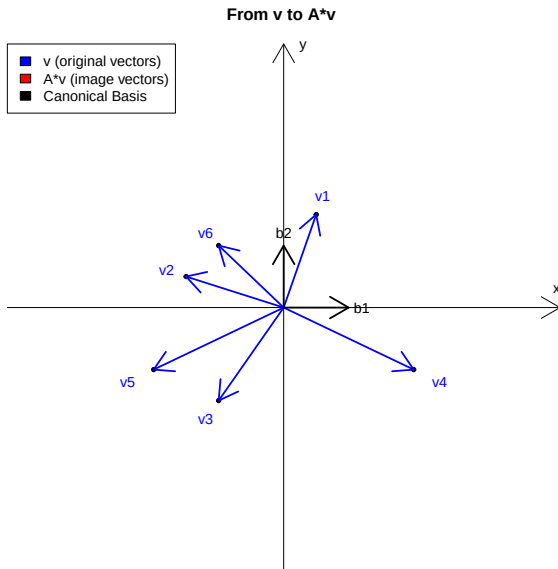
Let's compute the images of the  $v_i$  for the linear transformation given by  $A$ :

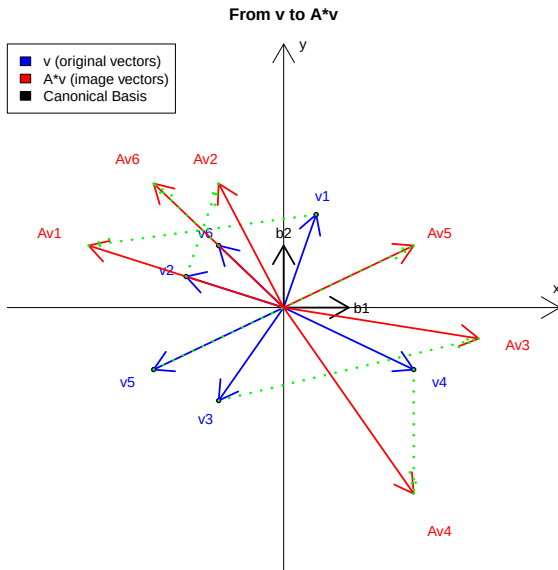
$$A = \begin{pmatrix} 0 & -2 \\ -1 & 1 \end{pmatrix}.$$

And the results are the following vectors in  $\mathbb{R}^2$

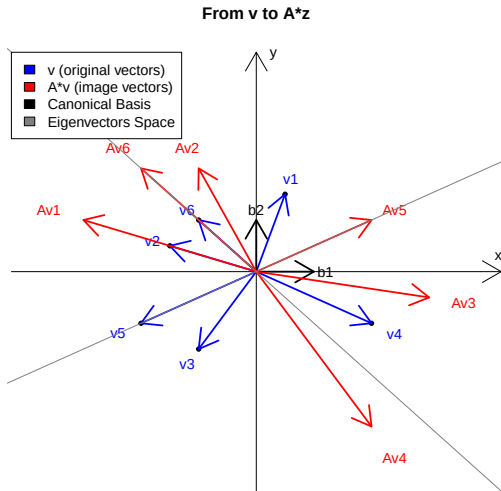
$$A.V_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix} \quad A.V_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad A.V_3 = \begin{pmatrix} 3 \\ -\frac{1}{2} \end{pmatrix}$$

$$A.V_4 = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \quad A.V_5 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad A.V_6 = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$$





# Surprise! Some Directions are Unchanged!



Unchanged directions= spaces of eigenvectors:

$$A.V_5 = A. \begin{pmatrix} -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} = -1. \begin{pmatrix} -2 \\ -1 \end{pmatrix} = -1.V_5$$

$$A.V_6 = A. \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} = 2. \begin{pmatrix} -1 \\ 1 \end{pmatrix} = 2.V_6$$

Values  $\lambda_1 = -1$  and  $\lambda_2 = 2$  are called eigenvalues and the vectors  $\vec{v}_5$  and  $\vec{v}_6$  are their respective eigenvectors.

These eigenvectors of an eigenvalue generate subspaces of eigenvectors:

$$E_{\lambda_1} = \left\{ \vec{w} = k. \vec{v}_5 \mid k \in \mathbb{R} \right\}$$

$$E_{\lambda_2} = \left\{ \vec{w} = k. \vec{v}_6 \mid k \in \mathbb{R} \right\}$$

If:

- $V$  is a vector space of dimension  $p$ ,
- $\mathcal{B} = \{\vec{b}_1, \dots, \vec{b}_p\}$ , is a basis of  $V$ ,
- $T : V \rightarrow V$  is a linear application on  $V$ ,
- $A \in \mathbb{R}^{p \times p}$  is the matrix representation of  $T$  in the basis  $\mathcal{B}$ ,
- for a vector  $\vec{v}$ , we denote  $V \in \mathbb{R}^{p \times 1}$  its coordinates in the basis  $\mathcal{B}$ ,
- $\lambda \in \mathbb{R}$ .

Then:

$\vec{v}$  is an eigenvector for  $T$  with eigenvalue  $\lambda$



$$T(\vec{v}) = \lambda \vec{v}.$$



$$AV = \lambda V.$$

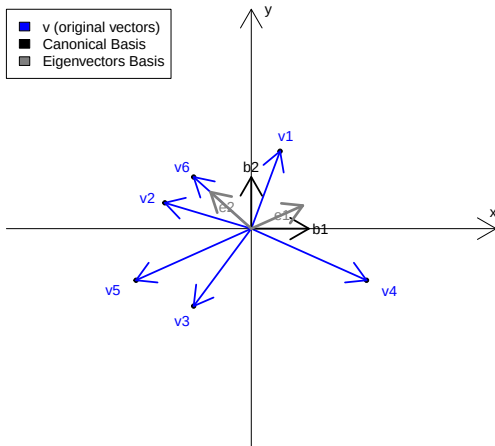


If we are lucky (i.e. if  $A$  is not singular), then set of eigenvectors can be used as basis of the space.

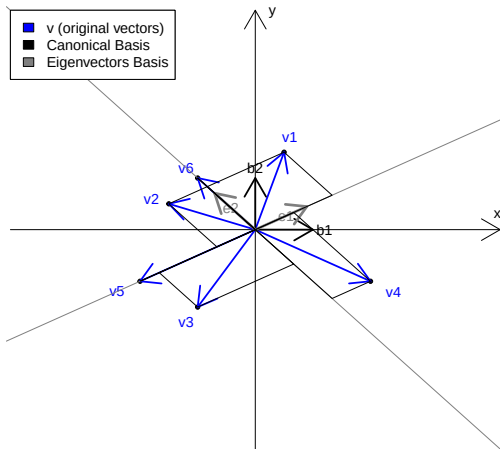
Then, any vector can be expressed using eigenvectors  $\{\vec{e}_1, \vec{e}_2\}$



Original Vectors and Eigen Basis



Original Vectors and Eigen Basis



It can be shown that

$$A = Q\Lambda Q^{-1}$$

where

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$$

and

$$Q = \begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix}$$

And if  $X$  are the coordinates of a vector  $\vec{v}$  in the canonical basis, then the coordinates  $Y$  in the eigenvectors basis are given by

$$Y = Q^{-1}.X$$

And finally the coordinates of the image  $T(\vec{v})$  in the eigenvectors basis are simply given

$$\Lambda.Y$$



Let  $\Lambda = \text{Diag}(\lambda_1, \dots, \lambda_p)$  the spectrum matrix of eigenvalues associated to  $A$  and  $Q = (q_1, \dots, q_p)$  a an eigenvector basis, each eigenvector of the basis being associated to an eigenvalue:

$$Q = \begin{pmatrix} q_1^1 & \cdots & q_1^j & \cdots & q_1^p \\ \vdots & & \vdots & & \vdots \\ q_p^1 & \cdots & q_p^j & \cdots & q_p^p \\ \updownarrow & \cdots & \updownarrow & \cdots & \updownarrow \end{pmatrix}$$

$$\Lambda = \begin{pmatrix} \lambda_1 & & & & \\ & \ddots & & & \\ & & \lambda_j & & \\ & & & \ddots & \\ & & & & \lambda_p \end{pmatrix}$$

- The “spectral” diagonalization of the matrix  $A$  is given by

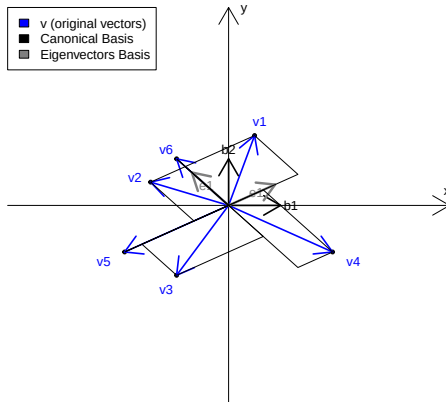
$$A = Q\Lambda Q^{-1}$$

- The matrix  $\Lambda$  is the matrix representation of the linear application  $T$  in the eigenvectors basis, columns of  $Q$ .
- Matrices  $Q$  et  $Q^{-1}$  are change-of-basis matrices which allow to change from canonical basis to the eigenvectors basis and vice-versa.
- We can prove that

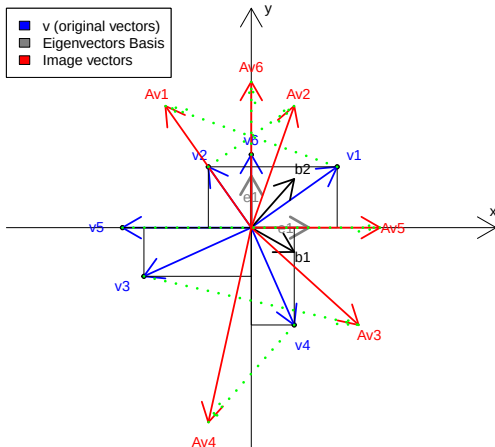
$$\text{Tr}(A) = \sum_{j=1}^p \lambda_j = \text{Tr}(\Lambda)$$



Original Vectors in the Space where  
Eigenvector Basis are Orthonormal Basis



Original Vectors in the Space where  
Eigenvector Basis are Orthonormal Basis



Let

- $V$  a vector space of dimension  $p$ ,
- $T : V \rightarrow V$  a linear application on  $V$ ,
- $A \in \mathbb{R}^{p \times p}$  the matrix representation of  $T$  in the canonical basis,
- $v$  is the eigenvector of  $A$  for the eigenvalue  $\lambda$  iff  $Av = \lambda v$ .

To find the eigenvalues et the eigenvectors for  $A$ :

- 1 find the eigenvalues  $\lambda$  solving the equation

$$\det(A - \lambda I_p) = 0$$

- 2 for each found  $\lambda$ , find the eigenvectors  $V$  for the given eigenvalue  $\lambda$  solving the system

$$AV = \lambda V.$$



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If we diagonalize the  $S$  variances-covariances matrix defined by

$$S = Z^t W Z$$

and if the diagonalization of  $S$  is given by

$$S = Q \Lambda Q^{-1}$$

then

- $Q$  is define  $p$  new variables, called components, linear combinations of original variables,
- the points of the original matrix  $X$  (and so the centered data matrix  $Z$ ) can be thus be represented in the basis vectors consisting of  $Q$ ,
- $\Lambda$  is the variance-covariance matrix of the components
- and like  $Tr(S) = \sum_{j=1}^p \lambda_j = Tr(\Lambda)$ , we have conservation of the total variance.



If the eigenvalues are sorted by decreasing values, i.e.

$\lambda_1 \geq \lambda_2 \geq \dots \lambda_{p-1} \geq \lambda_p$ , as we always have :

$$\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_p \end{pmatrix}$$
$$Q = \begin{pmatrix} \updownarrow & \dots & \updownarrow \\ q_1^1 & \dots & q_1^p \\ \vdots & & \vdots \\ q_p^1 & \dots & q_p^p \end{pmatrix}$$

We know that the component represented by vector  $q^1$  contains  $\lambda_1 / \text{Tr}(S)\%$  of the total variance...

And then components  $q^1$  et  $q^2$  contain  $(\lambda_1 + \lambda_2) / \text{Tr}(S)\%$  of the total variance...



We choose thus the  $k$  first components, most of the times such

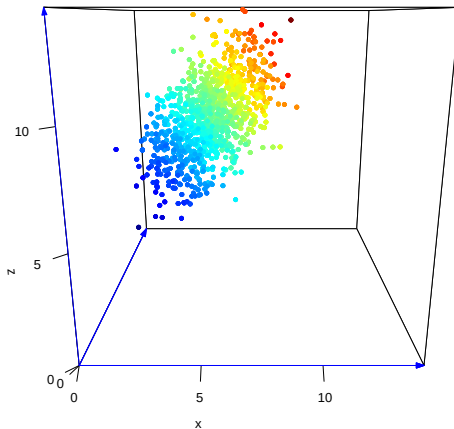
$$\frac{\sum_{j=1}^k \lambda_i}{\text{Tr}(S)} = \frac{\sum_{j=1}^k \sigma(x^j)}{\text{Tr}(S)} \geq p$$

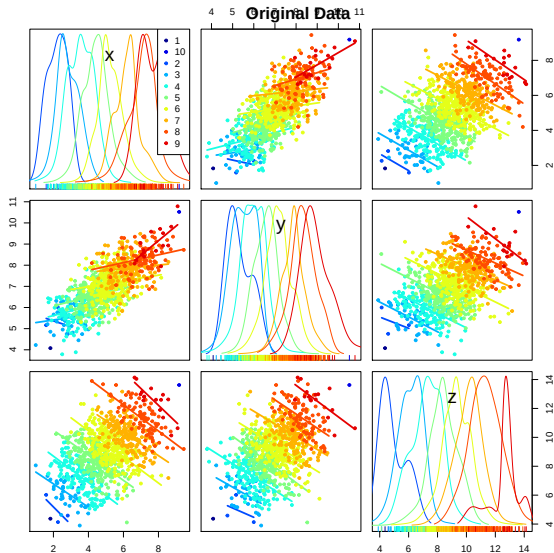
with  $p \in [0, 1]$ .

These components, with covariance matrix  $S$  are uncorrelated now!  
Remains to represent data in projected in the  $k$  main components, and the latter relative to the former basis ...



Original Data





The mean vector:

$$\bar{m} = \begin{pmatrix} 4.945 \\ 6.954 \\ 8.955 \end{pmatrix}$$

The covariance matrix:

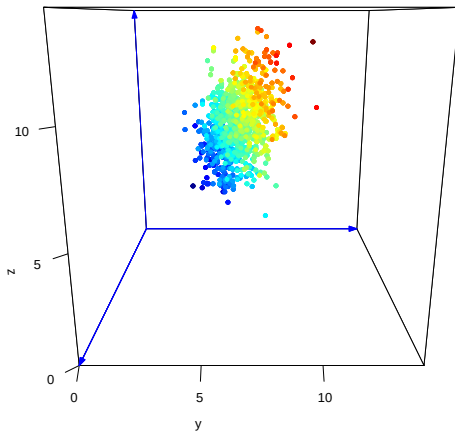
$$S = \begin{pmatrix} 2.035 & 1.255 & 1.029 \\ 1.255 & 0.998 & 0.728 \\ 1.029 & 0.728 & 3.188 \end{pmatrix}$$

The total Variance:

$$Tr(S) = 6.221$$



Centering Data



Spectral diagonalization of  $S$ :

$$S = Q\Lambda Q^{-1}$$

with

$$\Lambda = \begin{pmatrix} 4.302 & 0 & 0 \\ 0 & 1.761 & 0 \\ 0 & 0 & 0.1581 \end{pmatrix}$$

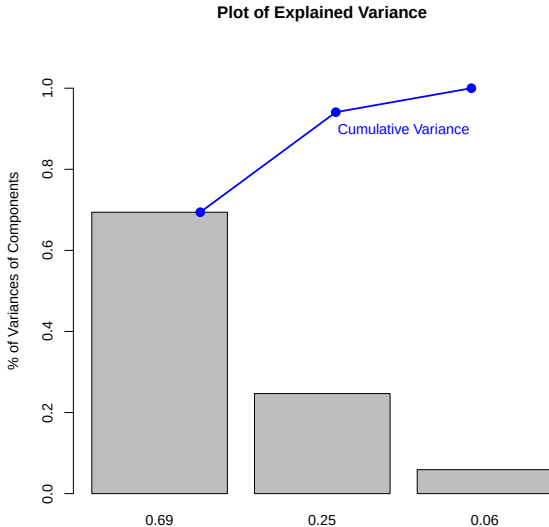
and

$$Q = \begin{pmatrix} -0.546 & 0.630 & 0.550 \\ -0.373 & 0.405 & -0.834 \\ -0.749 & -0.661 & 0.013 \end{pmatrix}$$

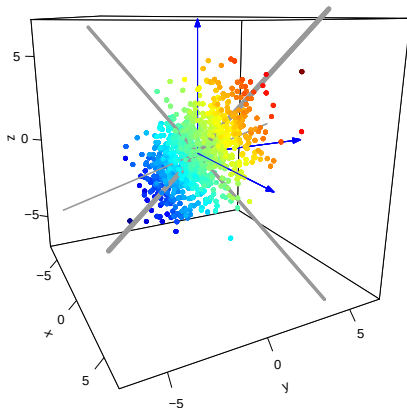
Note that

$$\text{Tr}(\Lambda) = 6.221 = \text{Tr}(S)$$





Eigenvectors Space  
Directions Sorted by Decreasing Variances



For each point of coordinates  $X$ , compute the new coordinates  $Y$  in the chosen components  $Y$  by:

$$Y = Q^{-1}X$$

And to show the canonical basis of the starting space in the components, for the coordinates  $B_i$  of each canonical vector  $\vec{b_i}$  compute its coordinates  $C_i$  in the components:

$$C_i = Q^{-1}B_i$$



